

Nb₄C₃T_x MXene — Multilayer Powder

Research-grade two-dimensional transition-metal carbide powder

Nb₄C₃T_x MXene is supplied as a multilayer powder derived from Nb₄AlC₃. The material is prepared using HF etching and retains a stacked two-dimensional layered structure. The grade is supplied for electrochemical, energy-storage, conductive-composite, interface and broader two-dimensional-material research.

KEY FEATURES

- Layered multilayer powder derived from Nb₄AlC₃
- Nominal flake / particle size: 2–20 μm
- Typical conductivity of 1–10 S/cm
- Surface terminations: –O, –OH, –F, –Cl

POTENTIAL RESEARCH APPLICATIONS

- Electrochemical and energy-storage material research
- Conductive-composite research
- Interface and two-dimensional-material research

STORAGE & HANDLING

Store dry in a tightly closed container. Vacuum or inert-atmosphere storage is recommended. Minimize air and moisture exposure.

QUALITY / DOCUMENTATION

Surface termination distributions vary with synthesis and lot. Quantitative surface composition can be discussed when required.

TECHNICAL SPECIFICATIONS

PROPERTY	TYPICAL / REPRESENTATIVE VALUE
Material	Nb ₄ C ₃ T _x MXene
Precursor	Nb ₄ AlC ₃
Product form	Multilayer Powder
Preparation	HF etching
Layer structure / count	Multilayer
Nominal flake / particle size	2–20 μm
Typical conductivity	1–10 S/cm
Surface terminations	–O, –OH, –F, –Cl
Recommended storage	Dry; vacuum or inert atmosphere

Conductivity values are representative technical values. Measurement results can vary with sample preparation, packing, film preparation and measurement geometry.

REPRESENTATIVE CHARACTERIZATION



Representative SEM morphology; not lot-specific.

WINIGEN MATERIALS

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Representative technical values. Properties can vary with synthesis and lot. Lot-specific documentation and additional analytical details are available subject to review.